T3 – Specification for Claude Code: fix duplicate customers in Ledger

1. Problem

The test migration created about 1,200 duplicate customers in the new CRM. We think the old CRM stores the same phone number in different formats (+91, leading 0, spaces), so the migration did not see them as one customer. Build a script that finds duplicates using a cleaned phone number, merges only clear matches, and sends doubtful cases to manual review. Never lose or hard-delete data. **Run on the TEST database only.**

2. Assumptions (confirm in section 7)

Python 3.11 + SQL (PostgreSQL). Table `customers(id, legacy_id, name, phone_raw, email, dob, city, created_at, updated_at, status, merged_into_id)`; child tables use `customer_id`. Customers are Indian unless a different country code is shown. A record may hold several phone numbers. Add columns `phone_e164` (cleaned) and keep `phone_raw` unchanged.

3. Phone cleaning – `normalize_phone(raw)` returns a list of "+91XXXXXXXXXX" or a flag

- Split on `/` `,` `;` or "or" and clean each number. Remove spaces, dashes, dots, brackets. Treat a leading `00` as `+`.
- Indian forms that all become the same 10 digits: `+91 9876543210`, `91-9876543210`, `0091 98765 43210`, `+91 0 9876543210`, `09876543210`, `98765 43210`. Rule: after cleaning, a 12-digit number starting 91 or an 11-digit number starting 0 loses that prefix; a 10-digit number is kept.
- **Valid mobile** = 10 digits, first digit 6–9, not all one digit, not `1234567890`. Other 10-digit numbers (start 1–5) = LANDLINE. Wrong length, letters, junk = INVALID. Both are never used for matching. A number with another country code (e.g. +44) is kept as given and matches only an exact copy.

4. Rules – when two records are the same customer

Name key: lower-case; remove Mr/Mrs/Ms/Dr/Shri/Smt, dots and commas; collapse spaces; sort words (so "Kumar Ravi" = "Ravi Kumar"). **Similar name** = one word differs by one character (typo), or a first name is only an initial matching the first letter of the other ("R Kumar" ~ "Ravi Kumar"); similar is never an exact match. **Conflict** = both records have a value and they differ (email or date of birth). A blank vs a value is not a conflict.

Rule	Condition	Action
R1	Same valid mobile (any of the record's numbers) AND same name key AND no conflict	**AUTO-MERGE**
R2	Same valid mobile AND (similar name, e.g. initials or one typo, OR same name with a conflict)	**MANUAL REVIEW** (reason CONFLICT or NAME_SIMILAR)
R3	Same email, similar or same name, different or invalid phone	**MANUAL REVIEW** (EMAIL_ONLY)
R4	Same valid mobile but clearly different names and no shared email/date of birth (family or shared phone)	**KEEP SEPARATE**; list in shared_phone.csv
R5	Phone invalid, landline or blank	Never match on phone; list in invalid_phone.csv
R6	Groups of 3+: merge only a group where *every pair* meets R1 (no chain merging); other members go to review	As above

Merge: survivor = record with most filled fields; tie = earliest `created_at`. Fill the survivor's blanks from the others; on different values keep the survivor's and store the others in the log. Re-point all child records to the survivor. Set the others to `status='MERGED'`, `merged_into_id` = survivor (soft delete). **Safety and audit:** default `--dry-run` (no writes); `--apply` required and refused unless the database name contains "test"; one transaction per group; running twice changes nothing; tables `merge_log` (run_id, rule, survivor_id, merged_ids, before/after JSON, time, operator) and `review_queue` (group, record ids, reason); CSV reports auto_merged, review_queue, shared_phone, invalid_phone; summary must satisfy records_before = active_after + merged.

5. Acceptance criteria

- **AC1** Given 3 records with the same name and phones "+91 9876543210", "09876543210", "98765 43210", When the script runs with `--apply`, Then one survivor stays ACTIVE and 2 become MERGED into it.
- **AC2** Given same phone and different names, When it runs, Then both stay ACTIVE and appear in shared_phone.csv.
- **AC3** Given same phone and same name but different email or date of birth, When it runs, Then nothing is merged and the group is in review_queue as CONFLICT.
- **AC4** Given a landline, junk or too-short phone, When it runs, Then it is not used for matching and is listed in invalid_phone.csv.
- **AC5** Given no --apply flag, When it runs, Then the database is unchanged and the reports show what would happen.
- **AC6** Given a merge, When it finishes, Then no child record points to a MERGED customer, merge_log has before/after data, and the totals reconcile.
- **AC7** Given the script has already run, When it runs again, Then it makes 0 changes.

6. Test data (expected result per case)

#	Input records (name; phone; other)	Expected result
1	A "Ravi Kumar"; +91 9876543210; r@x.com – B "Ravi Kumar"; 09876543210; no email – C "ravi kumar"; 98765 43210; r@x.com	R1: A, B, C merge into oldest; email kept
2	"Anita Sharma"; 91-9123456780; a@x.com; DOB 1985-02-11 – "Anita Sharma"; 9123456780; b@y.com; DOB 1985-02-11	Not merged; review (CONFLICT)
3	"Suresh Patel"; 9988776655; DOB 1970-01-01 – "Meena Patel"; +91 99887 76655; DOB 1974-03-09	R4: both stay; shared_phone.csv
4	"Kiran Rao"; 0091 98765-00011 – "Kiran Rao"; +919876500011	R1: merge
5	"Test User"; 0000000000 – "Test User"; 0000000000	Not merged; invalid_phone.csv
6	"Priya Nair"; 98765 4321 (9 digits)	Untouched; invalid_phone.csv
7	Case 1 records plus D "R. Kumar"; 9876543210	A, B, C merge; D to review (NAME_SIMILAR)
8	"Dev Singh"; +91 9812345678; dev@x.com – "Dev Singh"; 9811111111; dev@x.com	Review (EMAIL_ONLY)
9	"Mohan Das"; 9876501234 / 9123456789 – "Mohan Das"; 09123456789	R1 on the second number: merge
10	"Joan Dsouza"; +44 7700 900123 – "Joan Dsouza"; 07700 900123	Not merged: different country codes, no rule applies (see question 4)

7. Questions for the client before work starts

1. Can two real customers share a phone (family, office)? How common? **2.** Which fields are reliable: email, date of birth, address, old CRM customer ID? **3.** Which record should survive (oldest, newest, most complete)? Any VIP or subscription priority? **4.** Are all numbers Indian? Any landlines or foreign numbers? **5.** Who reviews the manual queue, and by when? What volume is acceptable? **6.** May merged records ever be deleted, or only hidden? Any retention rule? **7.** Are some of the 1,200 true duplicates already in the old CRM, and should they also be merged?